

K2-30b

R-2206

Benchmark run · workflow W8-benchmark-deep · record deep-bench_K2-30_250107 · created 2026-08-14T01:10:37+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460682.786 +/- 0.014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.163 +/- 0.047
Transit depth (Rp/Rs)^2	2.66 +/- 1.55 [%]
Orbital Inclination (inc)	89.93 +/- 2.3
Airmass coefficient 1 (a1)	0.646 +/- 0.08
Airmass coefficient 2 (a2)	0.35 +/- 0.12
Scatter in the residuals of the lightcurve fit is	14.34 %
Best Comparison Star	#1 - [387, 442]
Optimal Aperture	2.81
Optimal Annulus	9.19
Transit Duration (day)	0.097 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.163 $\pm$ 0.047 vs 0.1266 (deviation 0.61 σ)
Duration fit vs published	0.097 d vs 0.0971 d (deviation 0.0 σ)
Mid-time O-C	33.7 min
Depth SNR	2.7
Residual scatter	14.34 %

Light curve

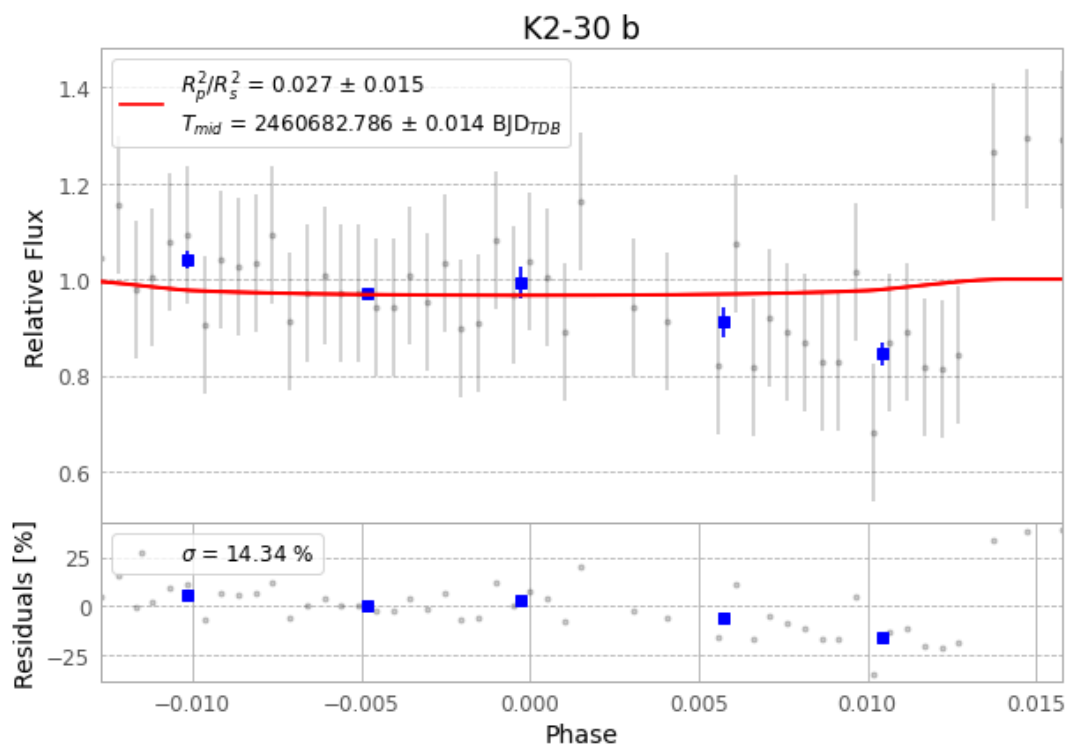


Figure 1 — FinalLightCurve_K2-30 b_2025-01-06.png (SHA-256 4155b9ef090d5f69...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [368, 276], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 891dbac011d0d6f7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

59 FITS frames (39 MB), K2-30250107053016.FITS → K2-30250107082415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest.

Artifacts: bench-K2-30-250107.log (8316b2ce45c8...), FinalLightCurve_K2-30 b_2025-01-06.png (4155b9ef090d...), FinalLightCurve_K2-30 b_2025-01-06.csv (119d5397500e...)

Compute resources

Wall time 174.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 01:10Z.